Experiment - 15

Error detection and correction using hamming code

Aim:

write a program to implement error detection and correction using hamming code

Create sender program

1) Input to sender file should be a text of any length. program should convert text to binary

2) Apply hamming code concept in binary data

3) save this output in file called channel

Create reciver program

1) reciver program should read the input from channel file

2) Apply hamming code on binary data

code:

```
data = list(map(int, input("Enter 4bits")))

p1 = data[0] ^ data[1] ^ data[3]

p2 = data[0] ^ data[2] ^ data[3]

p3 = data[1] ^ data[2] ^ data[3]

code = [p1, p2, data[0], p3, data[1], data[2],
        data[3]]
print("send code", code)

r = list(map(int, input("Enter received 7b

s1 = r[0] ^ r[2] ^ r[4] ^ r[6]

s2 = r[1] ^ r[2] ^ r[5] ^ r[6]

s3 = r[3] ^ r[4] ^ r[5] ^ r[6]

error = s1 + 2*s2 + 4*s3

if error:
    print("Error at bit", error)
    r[error-1] ^= 1
    print("Corrected:", r)
else:
    print("no error detected")
```

3) If there is an error, display the position of the error

4) else remove the redundant bits and convert the binary data to asii

Result: 13/X 10/10

Thus the program for error detection and correction was implemented successfully